

Parity Statistics of Nested Floor Powers

Four-Step Descent and Conditional Extensions for the Juggler Map

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Abstract

The Juggler map applies the integer part of the square root at even positive integers and of the three-halves power at odd positive integers. We prove that the starting values admitting a power-envelope descent certificate within four operations have natural density $13/16$. The analytic input is a mixed exponential-sum estimate for two nested floor powers and a smooth third coordinate. After one differencing, the remaining floor error is small in the required frequency range. An exact carry expansion and second-derivative estimates summed over every gap cell give the error $O(N^{23/24}(\log(2N))^3)$ for the four-step count. We also repair the OE-branch third-letter count using an exact square-root identity, and give a partition-aware curvature estimate for a basic collision model. These arguments restore part of an earlier draft without asserting its general decorated kernel bound. Under two additional finite-chain correlation hypotheses the five-step certificate density is $7/8$. Fair-share densities for every fixed-depth odd-rooted itinerary would imply density-one finite certificates; no uniformity in depth is needed for that implication. The general kernel estimate, its short-interval extension, and the remaining correlation hypotheses are unproved here. No result asserts universal arrival at 1.

Keywords: Juggler map; nested floor powers; parity correlations; discrepancy; stopping time; conditional descent.

1. Scope and relation to earlier work

For a positive integer n , define

$$J(n) = \begin{cases} \lfloor n^{1/2} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

This is the Juggler map recorded in OEIS A094683 [1]. The conjecture that every positive orbit reaches 1 remains open. The question studied here is more limited: which distribution estimates for finite itineraries would imply that many starting values eventually fall below their initial value?

Single-floor parity estimates follow from classical exponential-sum methods [2, 3]. Composing floors creates an additional difficulty. For $m = \lfloor n^{3/2} \rfloor$ and $\theta = \{n^{3/2}\}$,

$$m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4}).$$

The coefficient of the fractional part grows with n . A theorem for the smooth phase $n^{9/4}$ therefore does not by itself estimate the nested phase. Nor does a parity estimate over consecutive starting values automatically apply to a sparse set of orbit images.

The contribution of this note is an explicit reduction: the exact identities, finite certificate classification, sufficient analytic hypotheses, and conditional consequences are stated separately. The single-floor estimate and the probabilistic counting argument use standard methods. The latter is in the spirit of stopping-time arguments for the Collatz map, such as Terras [4]; no theorem about Collatz is transferred to the Juggler map.

This version replaces an earlier working draft and the first conditional revision. Section 4.2 repairs the restricted 23/24 estimate needed for four-step certificates by a complete small-shift argument. The general 95/96 kernel estimate remains open here. Section 7.2 repairs a basic collision model while stating the additional work needed for the decorated kernel. The former threshold $3.6 \cdot 10^{13}$ is not certified by this version: the analytic constants are implicit and no numerical threshold is claimed. The hypotheses for the higher conditional results remain explicit.

Throughout, $e(t) = \exp(2\pi it)$, $\{t\} = t - \lfloor t \rfloor$, and $\psi(t) = (-1)^{\lfloor t \rfloor}$. Natural density is measured among all positive integers. Thus a length- d odd-rooted word with count $2^{-d}N + o(N)$ has relative density $2^{-(d-1)}$ among odd starts. Constants in O and $\ll$ may depend on fixed words and explicit fixed parameters. They are not asserted to be uniform in depth.

2. Exact itineraries and the power envelope

Let $w = w_1 \cdots w_d$ be a word in $\{E, O\}$. The itinerary word $_d(n)$ records the parities of $n, J(n), \dots, J^{d-1}(n)$. It contains d letters and specifies the d operations that produce $J^d(n)$. Write $o(w)$ for the number of its odd letters.

Proposition 2.1 (power envelope). If w is realized at n , then

$$(J^d(n))^{2^d} \leq n^{3^{o(w)}}.$$

Consequently, if $n \geq 2$ and $3^{o(w)} < 2^d$, then $J^d(n) < n$.

Proof. At any positive integer x , the even operation satisfies $J(x)^2 \leq x$, and the odd operation satisfies $J(x)^2 \leq x^3$. Suppose a prefix of length r with o odd letters ends at x and obeys $x^{2^r} \leq n^{3^o}$. An even operation preserves the right-hand exponent after raising $J(x)^2 \leq x$ to 2^r ; an odd operation replaces it by 3^{o+1} . Induction starts at the empty prefix. If $3^{o(w)} < 2^d$ and $n > 1$, the bound is strictly less than n^{2^d} , proving descent. $\square$

A word satisfying $3^{o(w)} < 2^{|w|}$ is called *contracting*. A realized contracting prefix is a *power-envelope descent certificate*. Let

$$\mathcal{C}_d = \{n \geq 2 : \text{some contracting prefix of length at most } d \text{ is realized at } n\}.$$

Also set

$$\mathcal{C}_\infty = \bigcup_{d \geq 1} \mathcal{C}_d.$$

These sets refer to this particular sufficient certificate. Flooring can cause descent even when the envelope does not certify it. Therefore $\mathcal{C}_d$ need not equal the set of all starts that descend within d steps.

For an odd-rooted target word w , define its *formal chain* for every odd n , regardless of whether n realizes w :

$$x_1^w(n) = n, \quad \xi_t^w(n) = (x_t^w(n))^{p_t}, \quad x_{t+1}^w(n) = \lfloor \xi_t^w(n) \rfloor \quad (1 \leq t < d),$$

where $p_t = 3/2$ for $w_t = O$ and $p_t = 1/2$ for $w_t = E$. Set $s_t^w(n) = \psi(\xi_t^w(n))$, and put $\epsilon_t = 1$ for $w_{t+1} = E$, $\epsilon_t = -1$ for $w_{t+1} = O$.

Lemma 2.2 (parity and branch indicators). For real x , $\lfloor x \rfloor$ is odd exactly when $\{x/2\} \in [1/2, 1)$. For every odd n and odd-rooted word w ,

$$\mathbf{1}_{\text{word}_d(n)=w} = 2^{-(d-1)} \prod_{t=1}^{d-1} (1 + \epsilon_t s_t^w(n)). \quad (2.1)$$

Proof. Writing $x = \lfloor x \rfloor + \{x\}$ proves the first assertion, including the half-open endpoints. Each factor divided by two tests the parity of x_{t+1}^w . Until the first failed test, the formal chain agrees with the true orbit, by induction on t . A failed test makes the product zero and prevents realization of w . If all tests pass, the chains agree through the required letters and the product is one. The empty product covers $d = 1$. $\square$

The use of formal chains is essential: the product is defined on every odd input before any correlation or Fourier expansion is made.

3. An unconditional certificate density

Let $M(N) = \#\{n \leq N : n \text{ odd}\} = N/2 + O(1)$, and set

$$S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} \psi(n^{3/2}).$$

Theorem 3.1 (single-floor parity). One has

$$S_O(N) = O(N^{5/6}).$$

In particular,

$$\begin{aligned} \#\{n \leq N : \text{word}_2(n) = OO\} &= N/4 + O(N^{5/6}), \\ \#(\mathcal{C}_2 \cap [1, N]) &= 3N/4 + O(N^{5/6}). \end{aligned}$$

Proof. Write $n = 2r + 1$ and $g(r) = \frac{1}{2}(2r + 1)^{3/2}$. Then $g''(r) = \frac{3}{2}(2r + 1)^{-1/2}$. On a dyadic block with $r \asymp Q$, the classical second-derivative test [2] gives

$$\left| \sum_{r \asymp Q} e(hg(r)) \right| \ll h^{1/2} Q^{3/4} + h^{-1/2} Q^{1/4} \quad (h \geq 1).$$

The Erdős–Turán inequality [3, Chapter 2] bounds the unnormalized interval discrepancy by

$$\ll Q/H + \sum_{h=1}^H \frac{1}{h} \left| \sum_{r \asymp Q} e(hg(r)) \right| \ll Q/H + H^{1/2} Q^{3/4} + Q^{1/4}.$$

Taking $H = \lfloor Q^{1/6} \rfloor$ for sufficiently large Q gives $O(Q^{5/6})$. Lemma 2.2 identifies parity with membership in $[1/2, 1)$; summing the dyadic bounds proves the first assertion. Since $S_O = M - 2\#OO$, it also proves the count of OO .

The only minimal contracting words of length at most two are E and OE . The first class has count $\lfloor N/2 \rfloor$, and the second has count $M - \#OO = N/4 + O(N^{5/6})$. They are disjoint, and Proposition 2.1 proves the stated certificate count. $\square$

This argument estimates starting values in an interval. It makes no independence assertion about successive orbit parities.

Proposition 3.2 (the OE third letter). One has

$$\#\{n \leq N : \text{word}_3(n) = w\} = N/8 + O(N^{5/6} \log(2N)), \quad w \in \{OEE, OEO\}.$$

Proof. For every real $x \geq 0$,

$$\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor :$$

for an integer $r \geq 0$, the inequalities $r^2 \leq \lfloor x \rfloor$ and $r^2 \leq x$ are equivalent. Consequently, with $m = \lfloor n^{3/2} \rfloor$, one has $\psi(m^{1/2}) = \psi(n^{3/4})$ exactly.

On a dyadic block apply the two-dimensional Erdős–Turán–Koksma inequality to $(\{n^{3/2}/2\}, \{n^{3/4}/2\})$ at odd n , with cutoff $H = \lfloor P^{1/6} \rfloor$. For a nonzero integer mode (i, l) in this range, the second-derivative estimate gives

$$\left| \sum e\left(\frac{i}{2}n^{3/2} + \frac{l}{2}n^{3/4}\right) \right| \ll \begin{cases} |i|^{1/2}P^{3/4} + |l|^{-1/2}P^{1/4}, & i \neq 0, \\ |l|^{1/2}P^{3/8} + |l|^{-1/2}P^{5/8}, & i = 0. \end{cases}$$

For $i \neq 0$, dominance follows from $|l/i|P^{-3/4} \leq P^{-7/12}$. Using the product weights of Proposition 4.2, the discrepancy is

$$O(P/H + H^{1/2}P^{3/4} \log(2H) + P^{5/8} \log(2H)),$$

hence $O(P^{5/6} \log P)$. The two boxes selecting even m and either parity of $\lfloor n^{3/4} \rfloor$ have area $1/4$. Each therefore counts $P/8$ odd starts on the block, with the stated error. Sum dyadically. $\square$

4. Explicit correlation hypotheses

For a fixed odd-rooted word w of length d and nonempty $A \subseteq \{1, \dots, d-1\}$, define

$$R_{w,A}(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} \prod_{t \in A} s_t^w(n).$$

Hypothesis $H(w; \delta)$. For some fixed $0 < \delta < 1$, every nonempty A satisfies $R_{w,A}(N) = O_{w,A}(N^{1-\delta})$ as $N \rightarrow \infty$.

Hypothesis $H_0(w)$. The same correlations satisfy $R_{w,A}(N) = o(N)$, without a prescribed rate.

These are assertions about the formal chains, not assumptions that the true orbit is a sequence of independent coin tosses. Hypothesis $H(w; \delta)$ implies $H_0(w)$.

Proposition 4.1 (conditional word count). Under $H(w; \delta)$,

$$\#\{n \leq N : \text{word}_d(n) = w\} = 2^{-d}N + O_w(N^{1-\delta}). \quad (4.1)$$

Under $H_0(w)$, the error is $o(N)$.

Proof. Expand (2.1) and sum. The empty subset contributes $2^{-(d-1)}M(N) = 2^{-d}N + O(1)$. The other $2^{d-1} - 1$ terms are fixed signed multiples of $R_{w,A}$, and there are finitely many for fixed w . $\square$

For example, $H(OOEE; \delta)$ concerns the seven nonempty products of

$$\psi(n^{3/2}), \quad \psi(m^{3/2}), \quad \psi(v^{1/2}), \quad m = \lfloor n^{3/2} \rfloor, \quad v = \lfloor m^{3/2} \rfloor.$$

Proposition 4.1 then gives the count $N/16 + O(N^{1-\delta})$ needed for four-step certificates. Theorem 3.1 establishes the first estimate. Corollary 4.6 below establishes all seven for every $0 < \delta < 1/24$.

4.1 A sufficient Fourier criterion

For $\mathbf{k} = (k_1, \dots, k_{d-1}) \in \mathbb{Z}^{d-1}$, let

$$S_{w,\mathbf{k}}(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} e \left(\frac{1}{2} \sum_{t=1}^{d-1} k_t \xi_t^w(n) \right).$$

Proposition 4.2 (conditional Fourier-to-parity transfer). Fix $d \geq 2$, w , and $0 < \eta, \sigma < 1$. Suppose, uniformly over all nonzero integer vectors with $\|\mathbf{k}\|_\infty \leq \lfloor N^\eta \rfloor$, that

$$|S_{w,\mathbf{k}}(N)| \ll_w N^{1-\sigma}. \quad (4.2)$$

Then $H(w; \delta)$ holds for every $0 < \delta < \min(\eta, \sigma)$. The constants may depend on these fixed parameters.

Proof. Apply the multidimensional Erdős–Turán–Koksma inequality [3, Chapter 2, p. 116] to the points $\mathbf{y}_n = (\{\xi_1^w(n)/2\}, \dots, \{\xi_{d-1}^w(n)/2\})$. For any half-open axis-parallel box $B \subseteq [0, 1)^{d-1}$, its unnormalized discrepancy is

$$\begin{aligned} |\#\{n \leq N \text{ odd} : \mathbf{y}_n \in B\} - M(N)|B|| &\ll_d \frac{M(N)}{H} + \sum_{0 < \|\mathbf{k}\|_\infty \leq H} \frac{|S_{w,\mathbf{k}}(N)|}{r(\mathbf{k})}, \\ r(\mathbf{k}) &= \prod_{t=1}^{d-1} \max(1, |k_t|). \end{aligned}$$

The weight sum is $O_d((1 + \log H)^{d-1})$. With $H = \lfloor N^\eta \rfloor$, (4.2) gives box discrepancy $O_w(N^{1-\eta} + N^{1-\sigma}(1 + \log N)^{d-1})$. Each nonempty sign product is constant, with value 1 or -1 , on the 2^{d-1} boxes obtained by splitting each coordinate at $1/2$. Its integral over the unit cube is zero. Summing their discrepancies therefore bounds $R_{w,A}$; absorbing the fixed logarithmic power proves the assertion. $\square$

The criterion needs every indicated mixed mode, including vectors with zero coordinates and mixed signs. A bound for one undecorated kernel is not a substitute for (4.2). Qualitatively, cancellation $S_{w,\mathbf{k}}(N) = o(N)$ for each fixed nonzero $\mathbf{k}$ also implies $H_0(w)$: keep H fixed in the displayed inequality, let $N \rightarrow \infty$, and then let $H \rightarrow \infty$.

4.2 A nested estimate sufficient for four-step descent

The general decorated kernel is not needed for four-step certificates. We prove the smaller mixed family directly. Constants in this subsection may depend on a fixed $C \geq 1$, but are uniform in all integer modes in their stated ranges.

Lemma 4.3 (a truncated carry expansion). Put

$$b(t) = \{t\} - \frac{1}{2}, \quad b_R(t) = - \sum_{1 \leq |r| \leq R} \frac{e(rt)}{2\pi i r}, \quad E_R(t) = \min \left(1, \frac{1}{R\|t\|} \right),$$

where $\|t\|$ denotes distance to the nearest integer, $E_R(t) = 1$ at integers, and $R \geq 2$ is integral. Then $b(t) = b_R(t) + O(E_R(t))$, including at integers. For $X(x) = x^{3/2}$, any interval $I \subseteq [P, 3P]$ satisfies

$$\sum_{\substack{n \in I \\ n \text{ odd}}} E_R(X(n)) \ll \frac{P \log(2R)}{R} + P^{5/6}. \quad (4.3)$$

Proof. Away from integers, the Fourier series of b is the displayed series with infinite range. Summation by parts bounds its tail by $O((R\|t\|)^{-1})$. If $\|t\| \leq R^{-1}$, pairing the modes r and $-r$

and using $|\sin(2\pi r t)| \leq 2\pi r \|t\|$ bounds the finite sum by an absolute constant. At an integer the finite sum is zero and $b(t) = -1/2$, so the asserted bound remains valid.

The proof of Theorem 3.1, with phase $(2r+1)^{3/2}$, gives interval discrepancy $O(P^{5/6})$ for the fractional parts $\{X(n)\}$ with odd $n \in I$. In particular the number within distance $z \leq 1/2$ of an integer is $O(Pz + P^{5/6})$. Split these distances at $R^{-1}, 2R^{-1}, 4R^{-1}, \dots$. Each resulting weighted count contributes $O(P/R + 2^{-j}P^{5/6})$, proving (4.3). This is a bound in terms of the ambient scale P ; it does not rescale a global exceptional set by the length of I . $\square$

Lemma 4.4 (small-shift nested sum). Write $m(n) = \lfloor n^{3/2} \rfloor$, $Y(n) = m(n)^{3/2}$, and $\Delta_h f(n) = f(n+2h) - f(n)$. If

$$1 \leq h \leq P^{1/12}, \quad 1 \leq |j| \leq CP^{1/24}, \quad |i|, |k| \leq CP^{1/24},$$

where i, j, k, h are integers, then

$$\left| \sum_{\substack{P < n \leq 2P-2h \\ n \text{ odd}}} e\left(\frac{i}{2}\Delta_h X(n) + \frac{j}{2}\Delta_h Y(n) + \frac{k}{2}\Delta_h(n^{9/8})\right) \right| \ll_C P^{7/8}(1 + h^{1/2}). \quad (4.4)$$

Proof. Conjugate the whole sum if necessary and put $u = j/2 > 0$. Thus $u \geq 1/2$; integrality of u is not required. Set $\theta = \{X(n)\}$, $\delta = \Delta_h X(n)$, and $g = m(n+2h) - m(n)$. Lemma 7.1 gives the exact identity

$$\Delta_h Y = A_h + \frac{3}{2}g(n+2h)^{3/4} - \frac{3}{2}\theta\Delta_h(n^{3/4}) + \Delta_h E, \quad (4.5)$$

where

$$A_h(x) = \frac{3}{2}x^{3/2}\Delta_h(x^{3/4}) - \frac{1}{2}\Delta_h(x^{9/4}), \quad A_h''(x) = O(h^2P^{-7/4}).$$

For the derivative assertion, write $A_h(x) = x^{9/4}a(2h/x)$, where $a(z) = \frac{3}{2}((1+z)^{3/4} - 1) - \frac{1}{2}((1+z)^{9/4} - 1)$. The identities $a(0) = a'(0) = 0$ and bounded derivatives of $a(z)/z^2$ near zero give the estimate. No floor is differentiated.

Deleting the last two terms of (4.5) from the phase costs

$$O(uhP^{3/4} + uP^{1/4}), \quad (4.6)$$

because $\Delta_h(x^{3/4}) = O(hP^{-1/4})$, $E(x) = O(P^{-3/4})$, and $|e(t) - 1| \leq 2\pi|t|$.

Partition the interval into level sets of $G = \lfloor \delta \rfloor$. Since $\delta'(x) \asymp hP^{-1/2}$, there are $O(1 + hP^{1/2})$ cells; their total length is at most P . On a cell put $z = \delta - G$. It is monotone in $[0, 1)$, and exactly

$$g = G + \kappa, \quad \kappa = z + b(X(n)) - b(X(n+2h)) \in \{0, 1\}. \quad (4.7)$$

Define the two smooth phases on this cell by

$$F_{G,\epsilon}(x) = uA_h(x) + \frac{3u}{2}(G + \epsilon)(x+2h)^{3/4} + \frac{i}{2}\delta(x) + \frac{k}{2}\Delta_h(x^{9/8}), \quad \epsilon \in \{0, 1\}.$$

The exponential left after (4.6) is exactly

$$(1-z)e(F_{G,0}) + ze(F_{G,1}) + (b(X(n)) - b(X(n+2h)))(e(F_{G,1}) - e(F_{G,0})). \quad (4.8)$$

Apply Lemma 4.3 with $R = \lfloor P^{1/4} \rfloor$ to the two sawtooths. Their total error after the partition is $O(P^{5/6} + P^{3/4} \log P)$: the cells partition the same input set, and the multiplying difference of exponentials has modulus at most two.

On every cell $G + \epsilon \asymp hP^{1/2}$, and

$$F_{G,\epsilon}''(x) = -\frac{9}{32}u(G + \epsilon)(x+2h)^{-5/4} + O(uh^2P^{-7/4} + |i|hP^{-3/2} + |k|hP^{-15/8}). \quad (4.9)$$

The error is $o(uhP^{-3/4})$, uniformly in the stated ranges. The second-derivative estimate [2, Theorem 2.2] has scale $\lambda \asymp uhP^{-3/4}$ and a bounded curvature ratio. After $n = 2r + 1$ it has the same form for odd integers, with implicit constants. Summing over all cells, and using partial summation for the monotone weights z and $1 - z$, gives

$$\ll (uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8}. \quad (4.10)$$

Each nonzero Fourier mode adds either $rX(x)$ or $rX(x + 2h)$ to one of the phases $F_{G,\epsilon}$. Its curvature has size $|r|P^{-1/2}$, which dominates (4.9), since

$$\frac{uhP^{-3/4}}{|r|P^{-1/2}} \ll_C P^{-1/8} \quad (1 \leq |r| \leq R).$$

There is no curvature-collision regime in this proof. Summing the second-derivative estimate over all gap cells costs, per mode,

$$\ll |r|^{1/2}P^{3/4} + h|r|^{-1/2}P^{3/4}.$$

The Fourier weights are $O(1/|r|)$, so their total is $O(R^{1/2}P^{3/4} + hP^{3/4})$. Collecting the costs gives

$$\begin{aligned} &\ll_C (uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + P^{7/8} + hP^{3/4} \\ &\quad + uhP^{3/4} + uP^{1/4} + P^{5/6} + P^{3/4} \log P. \end{aligned}$$

Here $uh \ll_C P^{1/8}$ and $u \geq 1/2$. Every term is $O_C(P^{7/8}(1 + h^{1/2}))$. Endpoint cells are included; no estimate is extended across their boundaries. $\square$

Theorem 4.5 (restricted mixed exponential sums). For every fixed $C \geq 1$, uniformly over nonzero integer triples (i, j, k) with $\max(|i|, |j|, |k|) \leq CP^{1/24}$, one has

$$\left| \sum_{\substack{P < n \leq 2P \\ n \text{ odd}}} e\left(\frac{i}{2}n^{3/2} + \frac{j}{2}m(n)^{3/2} + \frac{k}{2}n^{9/8}\right) \right| \ll_C P^{23/24}. \quad (4.11)$$

Proof. If $j \neq 0$, apply the van der Corput differencing inequality [2] to the sequence indexed by odd integers, with $H = \lfloor P^{1/12} \rfloor$. Writing T_h for (4.4), it gives

$$|S|^2 \ll \frac{P^2}{H} + \frac{P}{H} \sum_{1 \leq h < H} |T_h| \ll_C \frac{P^2}{H} + P^{15/8}H^{1/2} \ll_C P^{23/12}.$$

If $j = 0$, the phase is smooth. For $i \neq 0$, its curvature is $\asymp |i|P^{-1/2}$, because $|k/i|P^{-3/8} \ll_C P^{-1/3}$. The bound is $O_C(P^{37/48} + P^{1/4})$. For $i = 0$, one has $k \neq 0$, curvature $\asymp |k|P^{-7/8}$, and bound $O_C(P^{7/12} + P^{7/16})$. Both are smaller than (4.11). $\square$

Corollary 4.6 (three formal signs). Put $v = \lfloor m^{3/2} \rfloor$. Every sign class of

$$(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$$

on odd $n \leq N$ has count

$$\frac{N}{16} + O(N^{23/24}(\log(2N))^3). \quad (4.12)$$

In particular, $H(OOEE; \delta)$ holds for every $0 < \delta < 1/24$.

Proof. On a dyadic block,

$$|v^{1/2} - n^{9/8}| \leq |v^{1/2} - m^{3/4}| + |m^{3/4} - X^{3/4}| \ll P^{-9/8} + P^{-3/8}.$$

Replacing the third phase coordinate in (4.11) by $v^{1/2}$ costs $O_C(|k|P^{5/8}) = O_C(P^{2/3})$. Apply the three-dimensional Erdős–Turán–Koksma inequality, as in Proposition 4.2, to the points

$$(\{X/2\}, \{Y/2\}, \{v^{1/2}/2\}),$$

with cutoff $\lfloor P^{1/24} \rfloor$ on this block. Its box discrepancy is $O(P^{23/24}(\log(2P))^3)$. The eight half-cube boxes encode the three signs exactly. Summing over dyadic blocks proves (4.12) and bounds all seven nonempty sign products by the same error. The cutoff is chosen separately on each block; no uniformity from a large block is assumed on smaller ones. $\square$

This establishes the hypothesis used for four-step certificates. It does not estimate the expanding third-level coordinate $v^{3/2}$ or the other formal chains required at length five.

5. Finite-depth descent densities

Lemma 5.1 (minimal certificates through length five). The contracting words of length at most five with no proper contracting prefix are exactly

$$E, \quad OE, \quad OOE, \quad OOOE, \quad OOEOE. \quad (5.1)$$

Proof. An E -rooted word already contracts at its first letter, and an OE -rooted word contracts at its second. Every remaining word begins with OO . At length three, even two odd letters give $3^2 > 2^3$, so no new certificate appears. At length four, contraction requires at most two odd letters, giving only $OOEE$. After excluding that prefix, the possible fourth-level prefixes are $OOEO$, $OOOE$, and $OOOO$. At length five, contraction requires at most three odd letters because $3^3 < 2^5 < 3^4$. The first two prefixes must therefore end with E , and the third cannot contract. This proves (5.1). $\square$

Theorem 5.2 (unconditional four-step density). One has

$$\#(\mathcal{C}_4 \cap [1, N]) = \frac{13N}{16} + O(N^{23/24}(\log(2N))^3).$$

Proof. Lemma 5.1 gives the disjoint prefix classes E , OE , and $OOEE$. Theorem 3.1 supplies the first two counts. The $OOEE$ class is the sign class $(-1, +1, +1)$ in Corollary 4.6, so has count $N/16 + O(N^{23/24}(\log(2N))^3)$. Their densities add to $1/2 + 1/4 + 1/16 = 13/16$. $\square$

Theorem 5.3 (conditional five-step density). Fix $0 < \delta_0 < 1/24$. Suppose $H(w; \delta_w)$ holds for each

$$w \in \mathcal{W} = \{OOOEE, OOEOE\}, \quad \delta_w > 0.$$

Then

$$\#(\mathcal{C}_5 \cap [1, N]) = \frac{7N}{8} + O(N^{1-\delta_*}), \quad \delta_* = \min(\delta_0, \min_{w \in \mathcal{W}} \delta_w).$$

The two qualitative hypotheses $H_0(w)$ suffice for natural density $7/8$.

Proof. The two additional prefix classes in (5.1) are disjoint from each other and from the four-step classes. Each has density $1/32$ by Proposition 4.1. Add these counts to Theorem 5.2, absorbing its logarithmic factor into $N^{1-\delta_0}$. The total density is $13/16 + 1/32 + 1/32 = 7/8$, and Lemma 5.1 exhausts certificates through length five. $\square$

The exact fractions and their analytic requirements can be read together:

Certificate length	Minimal prefixes added	Cumulative density	Status
1	E	$1/2$	Unconditional
2	OE	$3/4$	Theorem 3.1
4	$OOEE$	$13/16$	Unconditional: Theorem 5.2

Certificate length	Minimal prefixes added	Cumulative density	Status
5	$OOOEE, OOEEOE$	$7/8$	Conditional on the two hypotheses in Theorem 5.3

For the five-letter words, the four tested real images are, respectively, $(n^{3/2}, m^{3/2}, v^{3/2}, z^{1/2})$, with $z = \lfloor v^{3/2} \rfloor$, and $(n^{3/2}, m^{3/2}, v^{1/2}, u^{3/2})$, with $u = \lfloor v^{1/2} \rfloor$. Each hypothesis concerns all fifteen nonempty sign products of its four coordinates.

Separately, if $H_0(w)$ holds for every odd-rooted word of length four, all eight such classes have density $1/16$. This complete census is conditional as well; it is stronger than the three-sign estimate proved in Corollary 4.6.

6. The fixed-depth reduction to density-one certificates

Hypothesis FD. For every fixed $d \geq 1$ and every odd-rooted word w of length d ,

$$\#\{n \leq N : \text{word}_d(n) = w\} = 2^{-d}N + o_w(N).$$

No common error rate in d is assumed. The family $H_0(w)$ over all such words implies FD.

Theorem 6.1 (conditional density-one certificate theorem). Under FD, $\mathcal{C}_\infty$ has natural density one.

Proof. Put $p = \log 2 / \log 3 > 1/2$. A word with no contracting prefix through length d must in particular have at least pd odd letters at length d . All such words begin with O . For a uniformly chosen odd-rooted word of length d , the number of odd letters is distributed as $1 + B_{d-1}$, where B_{d-1} is a binomial random variable with parameters $(d-1, 1/2)$. This is an auxiliary count on words, not a stochastic model asserted for the Juggler orbit.

Choose $q = (p + 1/2)/2$, so $1/2 < q < p$. For all sufficiently large fixed d , $pd - 1 \geq q(d-1)$. For $t > 0$, the exponential Markov inequality and the binomial generating function give

$$\Pr(B_{d-1} \geq q(d-1)) \leq \left(\frac{1 + e^t}{2e^{qt}} \right)^{d-1}.$$

The logarithm of the expression in parentheses is zero at $t = 0$ and has derivative $1/2 - q < 0$ there. Hence some fixed $t > 0$ makes it a number $\vartheta < 1$.

For fixed d , FD and a finite sum over surviving words show that the natural density of $\mathbb{N} \setminus \mathcal{C}_d$ is their number divided by 2^d . It is at most $\frac{1}{2}\vartheta^{d-1}$ for sufficiently large d . Since $\mathcal{C}_d \subseteq \mathcal{C}_\infty$,

$$\overline{\text{dens}}(\mathbb{N} \setminus \mathcal{C}_\infty) \leq \frac{1}{2}\vartheta^{d-1}.$$

Letting $d \rightarrow \infty$ proves the assertion. In this argument $N \rightarrow \infty$ is taken first, with d fixed. $\square$

The conclusion concerns descent below the initial value. It does not prove that a descended value reaches 1, or that subsequent orbit samples avoid an exceptional set. Even a density-zero exceptional set can contain a nontrivial cycle. No interchange of the two limits, and no assertion of finite stopping time for every start, follows from Theorem 6.1.

7. Exact floor defects and the remaining kernel estimate

For $n \geq 3$, write

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m, \quad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v.$$

Lemma 7.1 (one-signed linearization). One has

$$m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + E(n), \quad 0 \leq E(n) \leq \frac{3}{8}(n^{3/2} - 1)^{-1/2} \leq \frac{1}{2}n^{-3/4}.$$

Proof. Let $a = \sqrt{m}$, $b = \sqrt{X}$. Direct factorization gives

$$E = a^3 - \frac{3}{2}a^2b + \frac{1}{2}b^3 = \frac{1}{2}(a - b)^2(2a + b) \geq 0.$$

Moreover $\theta = b^2 - a^2$, and $Ea \leq \frac{3}{8}\theta^2$ follows from $4a(2a + b) \leq 3(a + b)^2$, equivalent to $(5a + 3b)(a - b) \leq 0$. Thus $E \leq \frac{3}{8}m^{-1/2}$, which is at most $\frac{3}{8}(X - 1)^{-1/2}$. Finally $(X - 1)^{-1/2} \leq \frac{4}{3}X^{-1/2}$ for $n \geq 3$. $\square$

Lemma 7.2 (gap and carry). For an integer $h \geq 1$, set $\Delta_h X(n) = X(n + 2h) - X(n)$. Then

$$m(n + 2h) - m(n) = \lfloor \Delta_h X(n) \rfloor + \kappa_h(n), \quad \kappa_h(n) = \mathbf{1}_{\{X(n)\} + \{\Delta_h X(n)\} \geq 1} \in \{0, 1\}.$$

Proof. Expand $X(n) + \Delta_h X(n)$ into integer and fractional parts, and take its floor. The sum of the two fractional parts lies in $[0, 2)$. $\square$

For $1 \leq h \leq P/4$ and $n \in (P, 2P]$, differentiation gives $(\Delta_h X)'(n) \asymp hP^{-1/2}$. The level sets of $\lfloor \Delta_h X \rfloor$ consequently form $O(1 + hP^{1/2})$ intervals. Full intervals have length $\asymp P^{1/2}/h$; the two endpoint intervals can be shorter. A coefficient involving this floor is frozen only on these intervals. A derivative estimate proved there must be summed over the partition, or replaced by a separate global argument with controlled errors.

Lemma 7.3 (next-level defect). One has

$$v^{3/2} = m^{9/4} - \frac{3}{2}v^{1/2}\theta_2 - \mathcal{R}, \quad 0 \leq \mathcal{R} \leq \frac{3}{8}v^{-1/2}\theta_2^2.$$

Proof. Apply Taylor's theorem to $x^{3/2}$ between v and $v + \theta_2 = Y$. Its second derivative is $\frac{3}{4}x^{-1/2} \leq \frac{3}{4}v^{-1/2}$ on that interval, and is positive. Rearranging the Taylor formula gives the result. $\square$

These identities alone do not supply cancellation. The restricted estimate from the earlier draft,

$$\left| \sum_{\substack{P < n \leq 2P \\ n \text{ odd}}} e\left(\frac{i}{2}n^{3/2} + \frac{j}{2}m^{3/2}\right) \right| \ll_{\varepsilon} P^{23/24+\varepsilon}, \quad |i| \leq 2P^{1/24}, \quad 1 \leq |j| \leq 2P^{1/24}, \quad (7.1)$$

now follows from Theorem 4.5 with $k = 0$ and $C = 2$. The remaining kernel target, with $c_k(n) = \frac{3k}{4}n^{9/8}$, is

$$K_k(P) = \sum_{\substack{P < n \leq 2P \\ n \text{ odd}}} e(c_k(n)\theta_2(n)), \quad |K_k(P)| \ll_{\varepsilon} P^{95/96+\varepsilon}, \quad 1 \leq k \leq P^{1/24}. \quad (7.2)$$

The indices i, j, k are integers. Estimate (7.1) is unconditional in this revision; (7.2), with its printed uniformity, remains an **open target**. Even if (7.2) were established, its transfer to the mixed-mode families needed at length five would still require proof. Lemma 4.4 handles half-integer coefficients $j/2$ directly. This does not extend the frequency domain of the earlier general decorated lemma, whose separate applications remain unproved.

Short-interval variants require separate estimates. A bound for a transition set $\Omega \subset (P, 2P]$ of the form $|\Omega| \ll P^a$ implies only $|\Omega \cap I| \leq \min(|I|, O(P^a))$. It does not imply a bound proportional to $|I|/P$. Therefore the earlier localized-kernel threshold $|I| \geq P^{29/48+\delta}$ is not asserted here.

7.1 An unconditional estimate averaged over a shift

Proposition 7.4 (shift average). Let $L \geq 1$, let $A_1 < \dots < A_L$ obey $|A_t - A_s| \geq a|t - s|$ with $a > 0$, and let $x_1, \dots, x_L$ be real. For $S_\lambda = \sum_{t=1}^L e(A_t\{x_t + \lambda\})$,

$$\left| \int_0^1 |S_\lambda|^2 d\lambda - L \right| \leq \frac{4}{\pi} \frac{L}{a} (1 + \log L). \quad (7.3)$$

For every $0 < \eta < 1$, outside a set of shifts of measure at most η ,

$$|S_\lambda| \leq \left[\frac{L}{\eta} \left(1 + \frac{4}{\pi a} (1 + \log L) \right) \right]^{1/2}.$$

Proof. Expanding the square gives the diagonal term L . For $t \neq s$, the integrand $e(A_t\{x_t + \lambda\} - A_s\{x_s + \lambda\})$ is periodic in λ . Integrate over a period starting at one of its two fractional-part breakpoints. There is at most one other interior breakpoint, giving at most two intervals on which the phase is affine with slope $A_t - A_s$. On each interval the integral has modulus at most $1/(\pi|A_t - A_s|)$. Hence the off-diagonal contribution is bounded by

$$\frac{2}{\pi a} \sum_{t \neq s} \frac{1}{|t - s|} = \frac{4}{\pi a} \sum_{r=1}^{L-1} \frac{L-r}{r} \leq \frac{4}{\pi} \frac{L}{a} (1 + \log L).$$

For $L = 1$ the off-diagonal sum is empty. Markov's inequality applied to $|S_\lambda|^2$ gives the second assertion. $\square$

The estimate allows a small exceptional set of shifts. It gives no bound for a specified shift, including $\lambda = 0$ in (7.2).

7.2 Counting the gap cells in a collision estimate

The following lemma repairs the cell-counting problem for a fixed curvature model. It does not assert that every decorated phase in the earlier kernel proof meets its hypotheses.

Lemma 7.5 (curvature estimates over a partition). Partition an interval of length at most P into interval cells, so that every subinterval of length L meets $O(1 + \nu L)$ cells. On each cell let f be a real C^2 function; its values and derivatives may jump at cell boundaries. Suppose that a continuous reference function Λ satisfies $|\Lambda| \ll M$, and that, for $0 < s \leq 1$, both sets

$$\{|\Lambda| \leq sM\}, \quad \{sM < |\Lambda| \leq 2sM\}$$

are unions of $O(1)$ intervals, each set of total length $O(Ps)$. Assume

$$|f'' - \Lambda| \leq \rho M \quad \text{on every cell}, \quad 0 \leq \rho \leq 1/8, \quad 0 < M \leq 1, \quad MP^2 \geq 1.$$

All constants in these assumptions are fixed. Then, summing over integers or over odd integers with the same implicit-constant convention,

$$\left| \sum e(f(n)) \right| \ll PM^{1/2} + (1 + \nu P)M^{-1/2} + (P/M)^{1/3} + P\rho + 1. \quad (7.4)$$

Proof. Take $\tau = \max(4\rho, (MP^2)^{-1/3})$. If τ is bounded below by a positive absolute constant, the terms $P\rho + (P/M)^{1/3}$ already give the trivial bound $O(P)$. Otherwise discard $\{|\Lambda| \leq \tau M\}$, containing $O(P\tau + 1)$ integers. Split the remainder into dyadic bands $|\Lambda| \asymp sM$, with $\tau \ll s \ll 1$, and the outer set $|\Lambda| > M$ if necessary. Each band has total length $O(Ps)$ and meets $O(1 + \nu Ps)$ cells. On its intersection with a cell the curvature has one sign and size $\asymp sM$.

Apply the second-derivative estimate on every such intersection. The resulting costs at scale s are

$$O(PM^{1/2}s^{3/2} + M^{-1/2}s^{-1/2} + \nu PM^{-1/2}s^{1/2}).$$

The geometric sum over bands is $O(PM^{1/2} + M^{-1/2}\tau^{-1/2} + \nu PM^{-1/2})$. Our choice gives $M^{-1/2}\tau^{-1/2} \leq (P/M)^{1/3}$ and $P\tau \ll P\rho + (P/M)^{1/3}$, proving (7.4). One may assign boundary integers to either adjacent cell and estimate that cell up to its endpoint. $\square$

Proposition 7.6 (the basic frozen-gap collision model). Let $u \geq 1/2$, $1 \leq h \leq P^{1/8}$, $uh \leq P^{1/2}$, and let $w > 0$ satisfy $c_1 uh P^{-1/4} \leq w \leq c_2 uh P^{-1/4}$, for fixed $0 < c_1 < c_2$. With A_h as in Lemma 4.4 and $G(x) = \lfloor (x + 2h)^{3/2} - x^{3/2} \rfloor$, put

$$f(x) = uA_h(x) + \frac{3u}{2}G(x)(x + 2h)^{3/4} + wx^{3/2}.$$

Then

$$\left| \sum_{\substack{P < n \leq 2P \\ n \text{ odd}}} e(f(n)) \right| \ll_{c_1, c_2} (uh)^{1/2} P^{5/8} + (h/u)^{1/2} P^{7/8} + (uh)^{-1/3} P^{7/12} + h + P^{1/2}/h + 1. \quad (7.5)$$

Proof. On a gap cell the frozen-coefficient derivative is

$$f''(x) = -\frac{9}{32}uG(x)(x + 2h)^{-5/4} + \frac{3}{4}wx^{-1/2} + uA_h''(x).$$

Use the reference function

$$\Lambda(x) = -\frac{27}{32}uhx^{-3/4} + \frac{3}{4}wx^{-1/2}, \quad M = uhP^{-3/4}.$$

The identities $G(x) = 3h\sqrt{x} + O(1 + h^2P^{-1/2})$ and $A_h'' = O(h^2P^{-7/4})$ show, as a comparison of values,

$$|f'' - \Lambda| \ll M(h/P + (hP^{1/2})^{-1}).$$

This does not differentiate G . The gap-cell partition has density parameter $\nu \asymp hP^{-1/2}$.

To verify the sublevel conditions, factor $\Lambda(x) = \frac{3}{4}x^{-3/4}(wx^{1/4} - 9uh/8)$. The expression in parentheses is strictly increasing, with derivative $\asymp uh/P$ in the stated collision range. Hence $|\Lambda| \leq sM$ occupies length $O(PS)$. Moreover Λ' has at most one zero, so the sublevel sets and dyadic bands have a bounded number of components. Apply Lemma 7.5 with $\rho = O(h/P + (hP^{1/2})^{-1}) = o(1)$. Its terms give (7.5); the extra $M^{-1/2}$ is absorbed by $(h/u)^{1/2}P^{7/8}$. $\square$

Thus the transition cost need not be paid independently on every gap cell: a global sublevel argument counts how many cells can meet each curvature band. To use this in the complete kernel proof one must still verify the reference-function hypotheses, partition density, and weighted sums for all additional Fourier modes and decorations. Neither (7.4) nor (7.5) supplies that verification, and neither rescales a transition-set bound by an arbitrary short-interval length.

8. Evidence, availability, and remaining work

The status of the principal statements is as follows.

Statement	Warrant in this version
Envelope, branch identity, minimal words, floor defects	Exact proofs in Sections 2, 5, and 7
Certificate density 3/4	Classical analytic proof in Theorem 3.1
Correlation-to-count and Fourier-to-parity implications	Proofs in Propositions 4.1 and 4.2
Certificate density 13/16	Unconditional: Corollary 4.6 and Theorem 5.2
Certificate density 7/8	Conditional: two five-letter hypotheses unproved
Density-one finite certificates	Conditional on FD; no depth uniformity assumed
Restricted mixed sums	Unconditional: Theorem 4.5
General decorated kernel and short-interval extension	Unproved; no numerical threshold certified

Statement	Warrant in this version
Basic frozen-gap collision model	Proposition 7.6; does not certify every decorated phase
Shift-averaged estimate	Direct integral proof in Proposition 7.4

Supporting software is available in the Balanced Ternary Mathematical Laboratory repository, <https://github.com/sneakyweasel/btlab>. The accompanying source package contains this manuscript, its LaTeX build, and an exact-arithmetic validation script. That script checks the finite certificate list, the indicator identity on a finite census, and selected algebraic inequalities. A second exact-arithmetic script checks the differenced identity, carry expansion, and exponent comparisons used in the repairs. Those checks support reproducibility; they are not proofs of asymptotic cancellation.

Earlier repository audits and Lean modules concern identities, inequalities, and arithmetic from the previous draft. This version claims no Lean formalization of its analytic hypotheses or of the complete manuscript. Earlier theorem numbers are superseded and should not be used as citations to unconditional nested estimates. Companion texts that rely on those estimates require their own review.

The unconditional certificate density proved here is now $13/16$. Reaching $7/8$ by this route requires the two five-letter correlation hypotheses in Theorem 5.3. Section 7.2 supplies a partition-aware estimate for a basic collision model, but all decorated phase families and their weights still need to be controlled before the former kernel theorem can be restored. A proof of all fixed-depth counts would imply density-one certificates by Theorem 6.1; a further argument would still be required to settle universal termination.

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Large language models were used throughout the development of the earlier draft and this revision, including the formulation and review of proof arguments, drafting, programming, and discussion of Lean formalizations. The September 2026 review with OpenAI Codex identified unresolved analytic steps and prepared this conditional formulation, its proofs, the subsequent four-step and collision-model repairs, and its release materials. AI assistance and automated checks are not independent mathematical validation. The author is responsible for the statements, proofs, code, and final approval of the preprint. The models are not authors.

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